

BlueProof: Community-Verified Mangrove Restoration Monitoring with Vision-Language Models and Conditional Mobile Payments

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Abstract—Kenya holds roughly 61,271 hectares of mangrove forest, storing some of the densest carbon stocks of any tropical ecosystem, yet the forest continues to be lost to cutting, urban encroachment and pest damage. Community-led blue carbon projects demonstrably work, but they have not scaled: the pioneering Kenyan project has issued only tens of thousands of credits in fifteen years. We argue that the binding constraint is not a shortage of willing communities but the cost and latency of monitoring, reporting and verification (MRV). This paper presents BlueProof, a system design in which community monitors submit geotagged plot photographs, a vision-language model (VLM) performs bounded corroboration of those submissions, deterministic satellite analysis provides canopy-scale context, and verified submissions trigger immediate conditional mobile-money payment. We make the architectural role of the VLM deliberately narrow. Recent evidence shows general-purpose VLMs degrade sharply on field imagery relative to curated photographs and fabricate species names at non-trivial rates, so BlueProof does not treat the model as an authoritative taxonomic oracle. Instead the model corroborates a monitor-declared species, assesses image legibility and plot condition, flags cutting and pest damage, and abstains into human review when confidence is low. We further argue for an incentive design that pays for verified *reporting* rather than for seedling survival, on the grounds that paying for outcomes creates a direct incentive to conceal mortality. We situate the design against the closest existing deployment, Ericsson’s Connected Mangroves in Malaysia, and argue that its premise transfers to Kenya while its apparatus does not: fixed environmental sensors address growing conditions under corporate sponsorship and volunteer labour, whereas the Kenyan constraint is attestation cost in the absence of a sponsor, with mobile money affording a payment loop the Malaysian model had no reason to build. We describe a reference implementation and, because no field data has yet been collected, we set out a pre-registered evaluation protocol with explicit metrics, baselines and falsification criteria rather than reporting results. We state the ecological, legal and equity limitations of the approach, including the risk that cheaper verification expands credit supply without improving what a community monitor earns.

Keywords—mangrove restoration, blue carbon, MRV, vision-language models, participatory monitoring, mobile money, Kenya

1. Introduction

Mangroves occupy a small fraction of the world’s forested area but store carbon at densities far above most terrestrial forests, buffer coastlines against storm surge, and support in-shore fisheries. In Kenya they cover approximately 61,271 hectares, of which about 35,678 hectares, some 62 percent, lie in Lamu County [1]. Measured total ecosystem carbon stocks in Lamu average $560.22 \pm 79.79 \text{ Mg C ha}^{-1}$, with 70.3 percent of that carbon held below ground in soil rather than in standing biomass [1]. That distribution matters for monitoring: most of the carbon at risk is invisible from above, and is released by disturbance of the sediment rather than only by removal of the canopy.

The forest is under sustained pressure. Between 1990 and 2019 Lamu alone lost 1,739 hectares, a decline of roughly 60 hectares per year, with associated emissions estimated at $30,840 \text{ Mg CO}_2\text{e yr}^{-1}$ [1]. Drivers differ by county. In Lamu and Kilifi the historical driver has been overharvesting for construction poles and fuelwood. Around Mombasa, and specifically in Tudor Creek and Port Reitz, urban encroachment and effluent are dominant. Pest damage to *Sonneratia alba*, a frontline species in the seaward zone, has emerged as a compounding threat, while *Xylocarpus granatum* is selectively removed for carving and construction.

Community-led restoration and protection is not an untested idea in this setting. Mikoko Pamoja at Gazi Bay,

Kwale County, began in 2010 and is generally described as the world’s first community-led blue carbon project, involving 1,081 participating households [4]. It works. What it has not done is scale: the project has issued on the order of 2.2×10^4 certified credits across roughly fifteen years [4]. Against a national estate of 61,271 hectares and a national loss rate measured in tens of hectares per year, that is a rounding error.

The natural question is why a demonstrably functional model has not replicated. Our position, which this paper formalises into a system design and an evaluation protocol rather than asserting as established fact, is that the binding constraint is the cost, latency and expertise-dependence of MRV. A recent systematic treatment of MRV in carbon farming finds that monitoring consumes the largest share of total MRV expenditure, that field sampling is the most labour-intensive component, and that cost burdens fall disproportionately on smaller operations to the point of threatening their economic viability [3]. The same work finds that digitalisation measures such as automated data entry can halve data-entry time, and concludes that policy support is likely required for MRV systems to be scalable for smaller participants [3]. A community forest association managing tens of hectares of creek-side mangrove faces precisely this structure: the verification machinery costs roughly what it costs regardless of project size, so small projects are priced out of the market that would pay them.

1.1. Contributions

This paper makes four contributions.

- (1) A system architecture for community mangrove MRV that separates deterministic computation (vegetation indices, thresholds, payment rules, ledger integrity) from probabilistic perception (image interpretation, natural language), and makes that boundary explicit and auditable.
- (2) A deliberately *bounded* role for a vision-language model, motivated by published evidence on VLM failure modes in field imagery, in which the model corroborates rather than determines species, and abstains into human review under a calibrated confidence threshold.
- (3) An incentive design that pays community monitors for verified *reporting* rather than for seedling survival, with an argument for why outcome-linked payment is expected to corrupt the underlying data.
- (4) A pre-registered evaluation protocol with named metrics, baselines and falsification criteria, published in advance of data collection, together with an explicit statement of the ecological, legal and equity limitations of the approach.
- (5) A structured transferability analysis against the closest existing deployment, Ericsson’s Connected Mangroves in Malaysia, identifying which of its findings generalise to the Kenyan setting and which of its architectural choices are artefacts of a sponsorship and labour model that Kenya does not share.

We deliberately report no accuracy figures, no pilot outcomes and no cost savings. None have been measured. Section 6 specifies how they should be measured and what result would falsify the central claim.

2. Background and Related Work

2.1. Remote Sensing of Mangrove Extent and Condition

Satellite monitoring of mangrove extent is mature. Sentinel-2 provides multispectral imagery at 10 m ground sample distance for the visible and near-infrared bands with a revisit interval of roughly five days, and its time series have been used to map mangrove extent, track disturbance and characterise recovery trajectories [5, 6]. Multi-index approaches combining vegetation, water and soil indices with machine learning classifiers have been applied to mangrove dynamics assessment using Sentinel-2 imagery [12].

Two limitations bear directly on our design. First, resolution. At 10 m, a single pixel covers 100 m². This is adequate for detecting stand-level clearing and for tracking canopy closure across a restoration block, and wholly inadequate for observing whether an individual propagule planted three months ago is alive. Any claim that satellite data verifies planting survival at plot level is unsupportable. Second, cloud. Optical revisit frequency is nominal, not effective; the Kenyan coast has substantial cloud cover, and the usable optical record is considerably sparser than the five-day figure implies. Synthetic aperture radar, which penetrates cloud,

is the standard mitigation and is treated here as future work rather than as an implemented capability.

The consequence is a division of labour rather than a choice: satellite data supplies hectare-scale context and loss alerting, and ground observation supplies plot-scale truth. Neither substitutes for the other.

2.2. MRV Cost as the Scaling Constraint

The MRV literature supports treating verification cost as a first-order design variable. Vanderheyden et al. [3] decompose MRV cost into monitoring approach, project design parameters such as farm count and monitoring frequency, and protocol requirements, and find monitoring to be the dominant cost component with field sampling the most labour-intensive element. Their finding that digitalisation can halve data-entry time establishes that process automation yields real savings, while their conclusion that smallholder scalability likely requires policy support indicates that automation alone may be insufficient. We take both findings seriously: BlueProof is positioned as a cost-reduction instrument whose sufficiency is an empirical question, not a settled one.

2.3. Vision-Language Models for Field Species Identification

This subsection is the reason the system is designed the way it is.

A natural design for participatory monitoring is to let a general-purpose vision-language model identify what is in a submitted photograph. Recent evidence argues strongly against relying on that capability without guardrails. Zhou et al. [2] evaluate general-purpose VLMs in the 2–8B parameter range against a specialised model on a 96-species identification task. They report three findings that constrain any system of this kind. Every model evaluated, general-purpose or specialist, degrades sharply when moving from curated photographs to field imagery, with domain gaps of 9.6 to 26.6 percentage points. The specialised model outperforms the general-purpose VLMs by 33.2 to 59.2 percentage points. And the general-purpose models fabricate species names at rates of 5.9 to 9.6 percent.

Notably, the authors characterise the degradation as reflecting general image legibility rather than a failure of fine-grained discrimination as such. That distinction is actionable. It suggests that a VLM may be more reliable when asked whether an image is legible and what condition is visible than when asked to commit to a species label, and that legibility assessment is itself a useful product of the model.

Three design consequences follow, and we adopt all three.

- **Do not make the VLM the taxonomic authority.** Species is declared by the monitor, who is standing at the plot and typically planted it, and the model corroborates or disputes that declaration. A dispute is a signal for review, not an automatic overwrite.
- **Make abstention a first-class output.** A fabrication rate near ten percent is unacceptable in a record that a carbon buyer will rely on. The system must be able to decline.
- **Leave room for a specialist model.** Where taxonomic precision is required, the architecture should admit a

domain-trained classifier of the kind that outperformed general VLMs in [2], with the VLM retained for the open-ended judgements it handles well.

2.4. Participatory Monitoring and the Expertise Bottleneck

Participatory and citizen-science monitoring has an established record in ecology, and the appeal in this setting is obvious: the people best positioned to observe a creek-side plot weekly are the people who live beside it and planted it. The recurring difficulty is not willingness or diligence but credibility. A carbon standard requires evidence that an auditor will accept, and the conventional route to that acceptance is expert presence, which is the precise cost that [3] identifies as dominant and as regressive against small projects.

This produces a bottleneck with a specific shape. Observation is cheap and abundant; attestation is expensive and scarce. Systems that respond by sending experts to observe do not scale. Systems that respond by simply trusting self-report do not clear the credibility bar. The interesting design space lies between: mechanisms that let abundant lay observation carry an attestation burden it could not carry alone, by adding independent corroboration and by making the incentive structure of the reporter inspectable. BlueProof is an attempt to occupy that space, and the two corroborating signals it adds are machine assessment of the submitted image and satellite observation of the surrounding canopy.

2.5. Gap

Existing work establishes that satellite monitoring of mangroves is feasible at canopy scale, that MRV cost is a dominant and regressive constraint, and that general VLMs are unreliable taxonomic oracles on field imagery. What is missing is a system design that accepts all three findings simultaneously: one that uses satellite data only for what it can support, uses a VLM only for what the evidence says it can do, and couples verification to a payment mechanism whose incentive structure does not corrupt the data being verified. BlueProof is a proposal for such a system.

3. Precedent and Transferability: Connected Mangroves

The proposition that networked technology can improve mangrove restoration outcomes is not speculative. It has been demonstrated at field scale in Malaysia. Because that deployment is the closest existing precedent to the present work, and because it differs from the Kenyan setting in ways that are instructive rather than incidental, we treat it as a comparative case rather than a citation.

3.1. The Malaysian Deployment

Ericsson launched the Connected Mangroves project in Selangor, Malaysia, in October 2015, working with a local technology partner, the Global Environment Centre and the community of Kampung Dato Hormat [7, 8]. Sensor systems deployed among planted saplings measure soil moisture, temperature and electrical conductivity together with ambient humidity and water level, relay readings over commercial cellular networks, and present them through a cloud dash-

board accessible to community leaders, non-governmental organisations, analysts and authorities, with automated alerts when thresholds are breached [7]. Each sensor installation covers roughly 2,500 m², on the order of 200 mangrove plants [7].

The reported results are meaningful. Against a regional baseline in which only about 40 percent of planted seedlings reach maturity, the project targeted a 50 percent improvement in survival, beginning with 200 seedlings in a first phase and scaling to 1,000 [7]. Subsequent reporting describes survival of approximately 80 percent across roughly 4,000 saplings planted in Malaysia, with the model replicated to a site in the Philippines [8].

3.2. What the Precedent Establishes, and What It Does Not

Connected Mangroves establishes three things that we adopt without reservation. Restoration survival is responsive to information: knowing that a plot is drying or inundated, in time to act, materially changes outcomes. Community custodianship is compatible with instrumentation. And a mangrove restoration technology programme can be sustained for the better part of a decade, which is longer than most pilots survive.

It does not, however, establish what BlueProof needs to establish, and the difference is not one of degree. Connected Mangroves is an *agronomic telemetry* system. It measures the abiotic conditions in which seedlings grow, so that people can intervene. It does not attest that a particular person performed a particular act of stewardship at a particular place and time, which is the artefact a carbon standard purchases and the artefact that community projects cannot currently produce cheaply. A soil moisture reading tells you the ground is dry. It does not tell you that Amina planted forty propagules on Tuesday, nor does it pay her for having done so.

The two systems therefore solve adjacent but distinct problems: one optimises growing conditions, the other manufactures verifiable evidence and couples it to compensation.

3.3. Points of Divergence

Table 1 sets out the structural differences that bear on transferability.

Four of these deserve elaboration.

The sponsorship question is decisive. Connected Mangroves is underwritten by a telecommunications vendor supplying hardware, connectivity and volunteer labour. That is a legitimate and generous arrangement, and it means the project's unit economics are not the project's own. A Kenyan community forest association has no standing equivalent. Any design that assumes donated hardware inherits a dependency that, if withdrawn, ends the programme. BlueProof therefore carries no per-site capital cost by construction, which is the strongest single argument for community photography over instrumentation in Table 2.

Sensor density does not scale to a national estate. At one installation per 0.25 hectare, covering Kenya's 61,271 hectares [1] would require on the order of 2.4×10^5 installations. Even the 42-hectare site considered in this work would

Table 1: Structural comparison between the Malaysian Connected Mangroves deployment and the constraints facing a Kenyan community mangrove project.

	Connected Mangroves, Malaysia	Kenyan coastal community project
Problem addressed	Seedling survival under variable conditions	Verifiable attestation for carbon finance
Primary sensing	Fixed abiotic sensors among saplings	Human observation of plot condition
Who bears the cost	Corporate sponsor supplies hardware and connectivity	Project must self-finance from verification value
Coverage density	One installation per ~ 0.25 ha	Must cover tens of hectares per monitor
Community role	Volunteer planting and basic maintenance	Paid monitoring as a livelihood
Payment loop	None; information is the output	Conditional payment on verified submission
Governance context	State forestry with long silvicultural tradition	Gazetted reserve, co-managed, contested use
Infrastructure advantage	Cellular IoT backhaul	Near-universal mobile money
Dominant failure risk	Sensor maintenance and sponsor withdrawal	Verification credibility and model error

require roughly 170. The Malaysian density is appropriate for an intensively studied demonstration plot and is not a national monitoring architecture. Human observers scale differently: a monitor already lives beside the creek, walks it anyway, and can cover hectares rather than fractions of one.

Kenya holds an infrastructure advantage the Malaysian project could not use. Mobile money is close to universal in Kenya, which makes it possible to settle a payment to an individual monitor within seconds of a verification passing. Connected Mangroves informs its stakeholders; it does not pay them per verified act, and in a volunteer model there is no reason it should. The conditional payment loop is not a refinement of the Malaysian design. It is a mechanism Kenya can build that Malaysia’s infrastructure and labour model did not call for, and it converts conservation from an activity that costs a community time into one that returns income.

The Malaysian success metric is a trap if imported directly. The headline result is survival rising from roughly 40 percent to roughly 80 percent [7, 8]. As a measure of ecological outcome that is exactly right. As a basis for paying people it is hazardous, for the reason set out in Section 4.7: once survival determines income, the reporter acquires an interest in the number. Connected Mangroves is insulated from this because its survival figures come from instruments rather than from paid reporters. A Kenyan system that pays monitors and simultaneously grades them on survival would not be insulated, and would corrupt the very statistic it was built to produce.

3.4. Implication for the Present Design

The transferable lesson from Malaysia is the premise, not the apparatus. Information improves restoration outcomes, and communities will sustain a technology programme over years. The apparatus does not transfer, because the Kenyan binding constraints are attestation cost, absence of a sponsor and the need for the work to pay, none of which fixed environmental sensors address.

BlueProof can therefore be read as the Malaysian premise reconstructed under Kenyan constraints: observation performed by people rather than instruments, because people are already present and instruments must be bought and maintained; verification performed by a model with a bounded and auditable role, because expert attestation is the scarce resource; and compensation settled immediately over mobile

money, because that is the infrastructure Kenya actually has and because a conservation system that pays the people doing the conservation is more likely to persist than one that does not.

4. System Design

4.1. Design Principles

P1: Separate the deterministic from the probabilistic. Vegetation index computation, cloud screening, confidence thresholds, payment arithmetic and ledger construction are ordinary code with defined behaviour. Image interpretation and natural language generation are probabilistic. The boundary is explicit in the module structure so that each side can be audited on its own terms. A recurring failure in applied machine learning for conservation is delegating arithmetic or geometry to a language model; we avoid it by construction.

P2: Abstain rather than guess. Given the fabrication rates reported in [2], the system treats low confidence as a routing decision, not a discount factor. Submissions below a configured confidence threshold, or whose assessed condition is *unclear*, are escalated to human review and trigger no payment.

P3: Pay for verified reporting, not for outcomes. This is discussed in Section 4.7.

P4: Degrade gracefully. Field monitors work in locations with intermittent connectivity and on inexpensive handsets. Every external dependency in the reference implementation has a deterministic offline mode, and the assistant falls back to rule-based responses if the model endpoint is unavailable, so that a monitor in the field is never left without an answer.

4.2. Architecture

Figure 1 shows the chain of custody from field observation to verified record.

4.3. Data Model and Chain of Custody

The schema encodes provenance explicitly. A *Site* carries a boundary polygon, an area and a baseline vegetation index. A *Plot* belongs to a site and carries a field code, a declared species, a planting date, a count of seedlings planted and a coordinate. A *MonitoringEvent* belongs to a plot and a monitor and carries both the monitor’s declaration and the model’s assessment as separate fields, so that agreement and

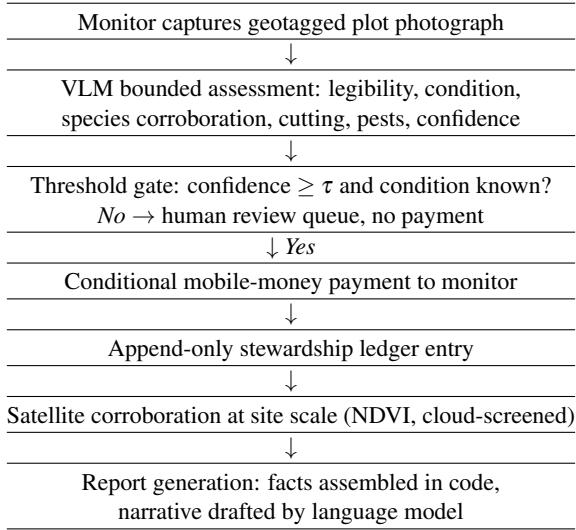


Figure 1: The BlueProof verification and payment pipeline. Probabilistic components are confined to the second and final stages; all gating arithmetic is deterministic.

disagreement remain recoverable after the fact. A *SatelliteObservation* belongs to a site and carries an index value, a change against baseline, a cloud fraction and an alert flag. A *Payment* belongs to a monitoring event. A *LedgerEntry* is written only after a payment settles and denormalises the verified facts into an append-only record: site, plot, species, count verified, condition, the named monitor, the amount paid and the coordinate.

Two properties of this schema matter for verification integrity. First, the monitor’s claim and the model’s assessment are never merged into a single field, so a later auditor can compute agreement rates and detect drift. Second, the ledger records the amount paid alongside the ecological facts, which makes the distribution of benefit inspectable rather than inferred. We return to why that matters in Section 7.2.

4.4. Bounded Role of the Vision-Language Model

The model is asked four questions about a submitted photograph, in this order of confidence in its answers:

- (a) **Is this image legible enough to assess?** Following [2], legibility is treated as the primary determinant of downstream reliability and is therefore assessed explicitly rather than implied by a confidence score.
- (b) **What condition is visible?** Healthy, stressed, dead or unclear. This is a coarse ordinal judgement over a visually salient property.
- (c) **Is there evidence of cutting or pest damage?** Stumps, cut poles and axe marks; boring holes, frass and dieback. These are presence detections of conspicuous features, and are treated as recall-oriented: a false positive costs a human review, a false negative loses a deforestation event.
- (d) **Is the monitor’s declared species consistent with the image?** Corroboration, not determination. Disagreement routes to review.

The model is given the nine mangrove species recorded in

Kenya as a closed label set, which reduces the space in which fabrication can occur, and any returned label outside that set is coerced to *unclear* rather than accepted. The architecture reserves a slot for a domain-trained classifier to replace the model on question (d), consistent with the specialist advantage reported in [2].

4.5. Satellite Corroboration

For each site, a normalised difference vegetation index series is computed from Sentinel-2 red and near-infrared bands over the site polygon and compared against a stored baseline. Observations whose cloud fraction exceeds a configured maximum are excluded from alerting rather than down-weighted, on the principle that an untrustworthy observation should not contribute to a decision that moves money or triggers an enforcement response. A sustained negative change beyond a configured threshold raises a canopy-loss alert for the site.

The satellite layer is explicitly *corroborative*. It cannot confirm plot survival, and the system does not claim that it does. Its value is threefold: it detects loss the field record might miss because nobody was sent to look, it provides an independent signal against which systematic misreporting would show up as divergence, and it supplies the landscape-scale context a report requires.

4.6. Alternative Sensing Modalities Considered

Community smartphone photography is not the only way to gather evidence from a mangrove stand, and the choice deserves justification rather than assumption. Table 2 compares four candidate modalities against the constraints that actually bind a small project in this setting.

Two of these deserve comment. Uncrewed aerial survey produces the most compelling imagery and is the weakest practical fit. In Kenya, drone operation requires registration supported by police clearance, and operation beyond the most restricted category requires a remote pilot licence together with operational authorisation, with commercial operation requiring an operator certificate; processing timelines are on the order of a month [11]. For a small project this converts a technical question into an administrative one, and an autonomous patrol falls squarely in the categories requiring prior authorisation. There is also a social cost that is easy to overlook: aerial cameras operating over a settlement read as surveillance regardless of intent, which is corrosive in a context where the system depends on voluntary participation.

Passive acoustic monitoring is scientifically attractive, particularly for faunal recovery, which is a genuine gap in vegetation-only assessment. It does not observe seedling survival, requires hardware deployment and retrieval, and depends on labelled regional audio that does not presently exist for this coastline. We regard it as a valuable complement at a later stage rather than a foundation.

The decisive argument for community photography is not sensing performance. It is that it is the only modality in Table 2 that transfers income to the community performing the conservation work. The observation and the livelihood are the same act. Every other modality treats the community as a subject of measurement rather than a participant in it.

Table 2: Sensing modalities for community mangrove monitoring, assessed against constraints binding on a small coastal project.

	Community photography	UAV survey	Passive acoustics	Fixed IoT sensors
Capital cost per site	None; existing handsets	High	Moderate per node	Moderate to high
Regulatory burden	None	Pilot licence, operator cert.	None	None
Realistic time to deploy	Days	Weeks to months	Weeks	Weeks
Field maintenance	None beyond the handset	Skilled operator	Retrieval, batteries	Power, theft, corrosion
Primary observable	Plot condition, disturbance	Canopy structure, extent	Faunal activity	Abiotic conditions
Plot-level survival	Yes	Partially	No	No
Community acceptance	High; participants are paid	Contested; surveillance framing	Neutral	Neutral
Generates livelihood	Yes	No	No	No

4.7. Conditional Payment

Payment is released only after a monitoring event passes the verification gate, and is delivered over mobile money, which in the Kenyan context means a near-instant transfer to a handset the monitor already owns.

The design decision that requires defending is what the payment is *for*. BlueProof pays a flat amount per verified submission. It does not pay per surviving seedling.

The argument is one of incentive compatibility. If payment is proportional to survival, a monitor who discovers that half a plot has died faces a direct financial penalty for reporting it accurately. The rational response is to photograph the surviving half, to report optimistically, or to replant before the assessment. Every one of those responses degrades exactly the data that the carbon buyer is purchasing, and the degradation is invisible precisely because the monitoring system has been trained on the resulting data. Paying for honest, verifiable reporting inverts this: mortality reported is still paid work, so the monitor’s interest and the buyer’s interest coincide. The verified record becomes worth something because it is not distorted by the mechanism that produced it.

This decision has a cost. It decouples payment from ecological outcome, which some funders will dislike, and it means that a monitor can in principle be paid for reporting a failed restoration. We consider that the correct trade: the alternative purchases a number that looks better and means less. We propose a field experiment to test the claim empirically in Section 6.

4.8. Report Generation

Monitoring standards require a structured narrative. In BlueProof the quantitative content of that narrative is assembled deterministically from the ledger and the satellite series, and the language model receives those figures as structured input with instructions to use only the supplied numbers, to state plainly where evidence is weak or contradictory, and to flag divergence between the field record and the satellite signal. The model is never asked to retrieve or estimate a quantity. If the model endpoint is unavailable, a deterministic template produces the same figures without narrative prose, so the report pipeline has no hard dependency on model availability.

5. Reference Implementation

A reference implementation exists and exercises the full pipeline end to end. The backend is a Python service built on FastAPI with SQLAlchemy over SQLite, chosen so that

Table 3: Service operations and their computational character. Probabilistic components are confined to three operations, all of which have deterministic fallbacks.

Operation	Character
Register site, plot	Deterministic
Assess photograph	Probabilistic; closed label set
Submit monitoring event	Deterministic gate over a probabilistic assessment
Release payment	Deterministic; gated
Write ledger entry	Deterministic; append-only
Refresh satellite series	Deterministic; cloud-screened
Generate report	Facts deterministic; narrative probabilistic
Field assistant	Probabilistic; rule-based fallback
Impact rollup	Deterministic

the system runs with no infrastructure provisioning during evaluation; the storage layer is portable to PostgreSQL, and the geometry representation to PostGIS, when spatial querying becomes necessary. The service exposes endpoints for site and plot registration, photograph assessment, monitoring submission with integrated verification and payment, satellite series refresh, report generation and an impact rollup, together with a monitor-facing assistant endpoint intended for a messaging channel.

Table 3 summarises the service surface and, for each operation, which side of the deterministic–probabilistic boundary it falls on.

Every external dependency has a deterministic offline mode: the vision model, the satellite source and the payment rail can each be independently switched to a live provider through configuration without code change. This is a testing convenience, but it is also a field requirement, since demonstrations and training sessions frequently happen without reliable connectivity.

The monitor-facing assistant answers practical questions in Swahili with some English, covering species zonation, spacing, how to record a plot and when payment is released. Zonation guidance follows the standard structure of Kenyan mangrove stands, with *Rhizophora mucronata* in regularly inundated lower zones, *Avicennia marina* in higher and more saline areas, *Ceriops tagal* intermediate, and *Sonneratia alba* seaward.

We report no performance measurements for this implementation. It has been exercised only against synthetic inputs.

6. Proposed Evaluation

Because no field data has been collected, we specify the evaluation in advance, including the results that would falsify the central claim.

6.1. Research Questions

RQ1. Can a general-purpose VLM, restricted to a closed set of nine Kenyan mangrove species, corroborate a monitor-declared species on field photographs at a rate useful for triage, and at what abstention rate?

RQ2. What is the model’s recall for conspicuous disturbance, namely cutting and pest damage, relative to an expert field assessment?

RQ3. Does the verification pipeline reduce MRV cost per hectare per monitoring cycle relative to current practice in a comparable community project?

RQ4. Does paying for verified reporting, rather than for survival, reduce misreporting of seedling mortality?

6.2. Dataset

We propose a corpus of approximately 1,000 geotagged plot photographs collected at Tudor Creek in Mombasa County and at Gazi Bay in Kwale County, the latter chosen because an established research baseline exists there. Images should be captured by community monitors on their own handsets, under the lighting and weather conditions that will obtain in deployment, rather than by researchers under favourable conditions; the domain gap reported in [2] is precisely the effect that a curated corpus would hide. Each image requires an independent expert label for species, condition and presence of disturbance, established by a trained botanist in the field at time of capture. Inter-rater agreement should be reported on a duplicated subset.

6.3. Metrics

- **Corroboration agreement:** rate at which the model’s species assessment agrees with the expert label, conditioned on the monitor’s declaration being correct and separately on it being incorrect. The second case is the operationally important one, since it measures whether the system catches a mistaken declaration.
- **Disturbance recall and precision,** with recall privileged. We propose reporting the precision obtained at a recall of 0.90 for cutting detection, since a missed clearing event is a materially worse error than an unnecessary human review.
- **Abstention rate and abstention quality:** the fraction of submissions routed to human review, and the error rate within the accepted set compared with the error rate within the abstained set. An abstention mechanism that does not concentrate errors in the abstained set is not working.
- **Fabrication rate:** frequency of species labels outside the closed set, or of asserted features absent from the image, measured against expert labels. This is a direct replication target for the 5.9 to 9.6 percent range reported in [2] under a constrained label set.

- **Calibration:** expected calibration error of the reported confidence against observed correctness, since the confidence threshold is the mechanism that gates payment and must be meaningful.

6.4. Baselines

Four comparators are required for the result to be interpretable: the monitor’s unaided declaration; a domain-trained specialist classifier of the kind shown to outperform general VLMs in [2]; an expert field officer, representing current practice and the cost baseline; and a trivial majority-class predictor, to establish the floor.

6.5. Cost Study

RQ3 requires an economic measurement, not a software benchmark. We propose recording, for one monitoring cycle at a comparable community project, the labour hours, travel, expert time and administrative overhead currently consumed per hectare, and comparing against the same quantities under the proposed workflow, inclusive of human review time generated by abstention and of model inference cost. Abstention is a cost transfer, not a cost elimination, and an evaluation that omits review labour would overstate the saving.

6.6. Incentive Experiment

RQ4 requires a field experiment with human participants and therefore ethical review, informed consent in the participants’ language, and the agreement of the relevant community forest association. We propose assigning monitoring groups to payment-for-reporting or payment-for-survival conditions and comparing reported mortality against independently assessed mortality on a sample of plots. The prediction is that reported mortality under payment-for-survival is significantly lower than independently assessed mortality, and that the gap is smaller under payment-for-reporting.

6.7. Threats to Validity

Construct validity. Expert labels are treated as ground truth, but species determination from a photograph is genuinely difficult for young individuals of morphologically similar taxa, and an expert working from the same image rather than from the plant in hand inherits the same limitation. We require labels established in the field at time of capture, and inter-rater agreement reported, so that the ceiling imposed by label noise is visible rather than assumed away.

Internal validity. Monitors who know their submissions are being evaluated may photograph more carefully than they would in routine operation, inflating measured legibility. A period of routine operation before the evaluation window, and sampling of submissions without prior notice, mitigate this only partially.

External validity. Two sites on one coastline, with one dominant set of species and one set of lighting and tidal conditions, will not characterise performance elsewhere. Results should be read as specific to Kenyan coastal mangrove and not generalised to other mangrove regions, still less to other ecosystems.

Temporal validity. Model behaviour changes as providers update endpoints. Any reported figure is a measurement of one model version at one point in time, and

the evaluation should record the exact model identifier and be treated as requiring periodic repetition rather than as a durable property of the system.

Conflict of interest. The evaluation of a cost-reduction claim is being proposed by the party who benefits from that claim being true. Cost measurement in particular should be performed or audited by the comparison project rather than by the system’s author.

6.8. Falsification Criteria

The central claim of this paper fails if any of the following hold. If corroboration agreement on genuinely mistaken declarations is no better than chance, the verification layer adds no value over self-report. If the abstention rate required to reach an acceptable error rate exceeds the fraction of submissions a human reviewer could process at the target scale, the system does not reduce expert dependence and the cost argument collapses. If the measured MRV cost per hectare is not materially below current practice once review labour is included, the premise of the work is wrong. We commit to reporting these outcomes if they obtain.

7. Discussion

7.1. What This System Does Not Do

BlueProof is not a carbon registry and does not issue credits. It produces evidence intended for use within an existing accredited methodology. Any sequestration figure derived from its outputs depends on a per-hectare rate that must come from an approved methodology or from site-specific measurement of the kind reported in [1], not from the system itself. The reference implementation carries such a rate as an explicitly labelled configuration assumption for demonstration purposes, and we would regard quoting a figure derived from it as a misrepresentation.

7.2. Equity and Benefit Sharing

There is a legitimate criticism of blue carbon finance that a limited share of revenue reaches the people performing the physical work. A system that reduces verification cost does not address this, and could plausibly worsen it: cheaper verification expands the supply of credits without altering how the proceeds are divided, and the efficiency gain may simply accrue to intermediaries.

We regard this as a design obligation rather than an externality. Two mitigations are built into the proposal. Recording the amount paid to a named monitor in the same ledger entry as the ecological facts makes the labour share of each verified record computable rather than obscured, and publishing the payment per verified event as a project parameter makes it contestable. Neither guarantees equitable distribution. Both make inequitable distribution visible, which is a precondition for contesting it. We would encourage any deployment to publish the ratio of monitor payments to total project revenue as a standing metric.

7.3. Where the Cost Actually Moves

It is worth being precise about the mechanism by which this design could reduce cost, because a vague claim of automation invites the reasonable objection that machine assessment

simply relocates work.

Current practice concentrates cost in expert field presence: travel to a site, time on the ground, and the administrative overhead of turning field notes into an auditable narrative. The proposed workflow leaves observation with people who are already present, which removes travel almost entirely for routine cycles, and replaces a portion of expert attestation with machine corroboration plus human review of the residual. Report preparation, which [3] associates with substantial administrative burden and where digitalisation is shown to yield measurable savings, is largely automated because the quantitative content is already structured.

Three components of cost do not reduce and must appear in any honest accounting. Human review of abstained submissions is real labour, and its volume is set by the abstention rate, which is in turn set by how conservative the confidence threshold must be to hold the error rate acceptable. Model inference has a per-submission cost. And initial training of monitors, together with periodic re-training, is unavoidable and is not a one-off. The cost study specified in Section 6 is constructed to capture all three, because an evaluation that counted only the removed travel would produce a flattering and false number.

7.4. Governance and Tenure

Mangroves in Kenya are gazetted government forest reserve. Any field deployment requires the participation of the Kenya Forest Service and a community forest association, and a system that generates a record of who did what where in a contested resource carries governance implications that software design cannot resolve. Photographs taken in and around a settlement are personal data in practice regardless of their classification in law, and retention, access and consent should be settled before collection begins, not after.

8. Limitations

We summarise the limitations stated throughout. The verification premise is untested on real mangrove photography, and the published evidence gives substantial reason for caution [2]. Satellite corroboration is implemented against a synthetic index series; live ingestion is specified but not built, and radar fallback for cloud is future work. The satellite layer cannot observe plot-level survival at 10 m resolution, and 70.3 percent of the carbon at stake in this ecosystem is in soil that no optical sensor observes [1]. The cost thesis is argued from the general MRV literature [3] rather than measured in this setting. The incentive argument is theoretical pending the experiment proposed above. And the system has been exercised only against synthetic inputs, so no claim about its behaviour under real load or real imagery is made here.

9. Conclusion

Kenya’s mangroves store carbon at exceptional density, are being lost at a measurable rate, and are protected in places by community institutions that work but do not scale. We have argued that verification cost is the plausible binding constraint, and presented a system design that attacks it while respecting two constraints the evidence imposes: that

satellite data cannot see what people claim it sees, and that general-purpose vision-language models are unreliable taxonomic authorities on field imagery. The resulting design confines the model to bounded judgements, requires it to abstain, keeps monitor claims and model assessments separately recorded, and pays for honest reporting rather than for favourable outcomes.

Whether this reduces MRV cost enough to matter is an empirical question that this paper does not answer. We have specified how to answer it, and what answer would tell us the idea is wrong.

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